Vilmodrine Film-Coated Tablets

Module 3 Quality Documentation — Formulation F-413/B

Process Validation Summary

Process performance qualification was executed on three consecutive commercial-scale batches. All predefined acceptance criteria for critical process parameters and critical quality attributes were met, supporting a conclusion of a validated state of control.

Cleaning verification swab results were below the health-based exposure limits derived for the product. Analytical results are trended electronically, and out-of-trend results trigger a documented laboratory investigation. The finished product is stored and distributed under controlled ambient conditions consistent with the approved label storage statement. Annual product quality reviews evaluate batch trends, deviations, complaints, and stability data to confirm the continued state of control. Hold-time studies established acceptable storage durations for intermediate blends and cores under defined conditions.

Change control procedures ensure that any modification to materials, methods, or equipment is assessed for regulatory impact prior to implementation. Supplier qualification includes periodic audits, review of certificates of analysis, and trending of incoming material test results. The control strategy links critical quality attributes to critical process parameters identified during development. Annual product quality reviews evaluate batch trends, deviations, complaints, and stability data to confirm the continued state of control. Training records for personnel involved in manufacturing and testing are maintained under the site quality system.

Container Closure System

The commercial packaging configuration consists of high-density polyethylene bottles with child-resistant polypropylene closures and induction seal liners. Container closure integrity was demonstrated by dye ingress testing. The packaging components comply with applicable food-contact regulations and compendial requirements for plastic packaging systems.

The finished product is stored and distributed under controlled ambient conditions consistent with the approved label storage statement. Environmental monitoring of the manufacturing areas is performed per the site monitoring program, with alert and action limits established from historical data. Analytical results are trended electronically, and out-of-trend results trigger a documented laboratory investigation. Data supporting these conclusions are maintained at the manufacturing site and are available for inspection. Cleaning verification swab results were below the health-based exposure limits derived for the product. Risk assessments were performed in accordance with ICH Q9 principles, and residual risks were judged acceptable.

Regional Information

No excipients of human or animal origin are used in the drug product with the exception of lactose monohydrate, for which a TSE/BSE compliance declaration is provided in Module 3.2.A.2. Certificates of suitability are maintained where applicable.

The finished product is stored and distributed under controlled ambient conditions consistent with the approved label storage statement. Data supporting these conclusions are maintained at the manufacturing site and are available for inspection. The control strategy links critical quality attributes to critical process parameters identified during development. Annual product quality reviews evaluate batch trends, deviations, complaints, and stability data to confirm the continued state of control. Supplier qualification includes periodic audits, review of certificates of analysis, and trending of incoming material test results. Deviations observed during the campaign were investigated and closed with no impact to product quality.

Control of Drug Product

The drug product specification includes tests for description, identification, assay, uniformity of dosage units, dissolution, related substances, water content, and microbial limits. Analytical procedures are compendial where available; non-compendial procedures were validated in accordance with ICH Q2(R1).

The finished product is stored and distributed under controlled ambient conditions consistent with the approved label storage statement. Analytical results are trended electronically, and out-of-trend results trigger a documented laboratory investigation. No changes to the approved analytical procedures are proposed in this submission. Cleaning verification swab results were below the health-based exposure limits derived for the product.

Manufacturing Process Overview

The drug product is manufactured by a conventional wet granulation process comprising dispensing, high-shear granulation, fluid-bed drying, milling, blending, compression, and film coating. In-process controls are applied at each unit operation, including loss on drying after fluid-bed drying and blend uniformity prior to compression. Process parameters were established during process performance qualification and are maintained under the site quality system.

Cleaning verification swab results were below the health-based exposure limits derived for the product. No changes to the approved analytical procedures are proposed in this submission. Change control procedures ensure that any modification to materials, methods, or equipment is assessed for regulatory impact prior to implementation. The finished product is stored and distributed under controlled ambient conditions consistent with the approved label storage statement. Environmental monitoring of the manufacturing areas is performed per the site monitoring program, with alert and action limits established from historical data. Risk assessments were performed in accordance with ICH Q9 principles, and residual risks were judged acceptable.

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3.2.P.4 Excipient Control Strategy

For the commercial formulation F-413/B, Kosher Passover grade HyQual (Magnesium stearate, lubricant) is used.

Grade Selection Rationale

Kosher Passover grade HyQual was selected as the commercial grade of magnesium stearate on the basis of lower moisture content mitigating hydrolytic degradation of the drug substance.

Grade selection and control follow the principles of FDA Guidance for Industry: Immediate Release Solid Oral Dosage Forms and ICH Q8(R2).

Registered Specification

The current approved commercial specification (SPEC-EX-4961 Rev 09) lists Magnesium stearate as Kosher Passover grade HyQual.

Regional Information

No excipients of human or animal origin are used in the drug product with the exception of lactose monohydrate, for which a TSE/BSE compliance declaration is provided in Module 3.2.A.2. Certificates of suitability are maintained where applicable.

Statistical evaluation of batch release data demonstrates process capability well within specification limits. No changes to the approved analytical procedures are proposed in this submission. Environmental monitoring of the manufacturing areas is performed per the site monitoring program, with alert and action limits established from historical data. Cleaning verification swab results were below the health-based exposure limits derived for the product. Hold-time studies established acceptable storage durations for intermediate blends and cores under defined conditions. The control strategy links critical quality attributes to critical process parameters identified during development.

Change control procedures ensure that any modification to materials, methods, or equipment is assessed for regulatory impact prior to implementation. The control strategy links critical quality attributes to critical process parameters identified during development. Environmental monitoring of the manufacturing areas is performed per the site monitoring program, with alert and action limits established from historical data. Annual product quality reviews evaluate batch trends, deviations, complaints, and stability data to confirm the continued state of control. Statistical evaluation of batch release data demonstrates process capability well within specification limits. Risk assessments were performed in accordance with ICH Q9 principles, and residual risks were judged acceptable.

Batch Formula

The representative commercial batch formula is provided for a nominal batch size of 400,000 tablets. Quantities of all components are expressed per tablet and per batch. Purified water is used as granulation fluid and is removed during processing; it does not appear in the final composition.

The control strategy links critical quality attributes to critical process parameters identified during development. Annual product quality reviews evaluate batch trends, deviations, complaints, and stability data to confirm the continued state of control. Risk assessments were performed in accordance with ICH Q9 principles, and residual risks were judged acceptable. Environmental monitoring of the manufacturing areas is performed per the site monitoring program, with alert and action limits established from historical data.